Table 2. EFFECT OF FRESHLY SPLIT SURFACES TOUCHING EACH OTHER

Dissipation factor (per cent) at 500 kc/s

Ruby			Muscovite	Green			Muscovite
Unsplit sample	Sample *A*	Sample *B*	Samples *A* and *B*, freshly split surfaces touching each other	Unsplit sample	Sample *C*	Sample *D*	Samples *C* and *D*, freshly split surfaces touching each other
0·017	0·052	0·034	0·045	0·137	0·215	0·186	0·131
0·052	0·098	0·098	0·069	0·079	0·160	0·111	0·118
0·034	0·074	0·052	0·065	0·114	0·158	0·239	0·114

into two pieces, and the dissipation factor of each piece was determined. Thereafter the two pieces were recombined with the fresh surfaces facing each other and the dissipation factor was redetermined. It is observed from Table 2 that after recombination the dissipation factor of ruby muscovite changed from the initial value of the unsplit sample, but higher than the value of one of the split specimens, whereas the green muscovite almost regained the original value of the unsplit specimen. On recombination, it would be expected that the charges on the two faces would neutralize each other and that the properties exhibited could be those of an unsplit mica. (Consideration would have to be taken of the fact that during the process of recombining some air bubbles might stick between the layers, thereby increasing the dissipation factor[5]. Furthermore, leakage of charge as well as evaporation of surface moisture during the process of recombining might also affect the result.)

A few samples were split in alcohol to a thickness of about 0·025 mm. The dissipation factor at 500 kc/s was found to be much higher (0·270 per cent) than for samples split in atmosphere (0·073 per cent). During splitting in alcohol it is possible that some of the OH ions adhere to the charged layer of the freshly split mica, thereby increasing the dissipation factor.

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GEOLOGY

Age and Polarity of the Turkana Lavas, North-west Kenya

THE extension of the Trap Series of Ethiopia into north-west Kenya has previously been suggested on stratigraphical and petrological grounds[1]. This area has recently been mapped on the basis of a reconnaissance[2], and the following succession suggested:

Rhyolites and andesites	*Tvr, Tvd*
Olivine basalts	Tvb_2
Phonolites and nephelinites	*Tvp*
Augite and analcime basalts	Tvb_1
Turkana grits	*Tm*

The age of the lavas has hitherto been based on a sparse mammalian fauna found in grits intercalated in basalts (Tvb_1 ?) at Lothidok, 25 km north-east of Lodwar. Initially thought Burdigalian[3], recent work[4] has suggested that it is younger (Up. Miocene) in age. Some fossil wood found in the rhyolites 60 km north of Lokitaung (*Dryoxylon* sp.) hints at a Mio-Pliocene age.

Four lavas have been dated (R. L. G.) at the University of Cambridge; Table 1 gives the results obtained on whole rock samples.

Field mapping showed that the basalts and rhyolites interdigitated in two localities, but neither were observed to overlie phonolites or nephelinites. The *Tvp* group is thus younger than both basalts (but could be older than the rhyolites), and comparable in age with the Lothidok grits and the Kapitian phonolites. The basalts (Tvb_1 ?) at

Table 1

Sites	K_2O (per cent)	Vol. of $^{40}Ar/g$ (mm^3)	Atmos. contam. (per cent)	Age (m. yr.)	Polarity
260/1	1·31	$1·406 \times 10^{-3}$	38·9	32·2 ± 0·5	*R*
(Tvb_1)		$1·375 \times 10^{-3}$	36·7	31·5 ± 0·7	
264/5	0·61	$4·210 \times 10^{-4}$	48·2	20·7 ± 0·4	*R*
(Tvb_1)		$4·686 \times 10^{-4}$	46·4	23·0 ± 0·7	
293/4	0·90	$7·04 \times 10^{-4}$	78·0	23·5 ± 1·1	*R*
(Tvb_2)		$6·99 \times 10^{-4}$	51·7	23·3 ± 0·2	
243/50	3·84	$1·90 \times 10^{-3}$	54·3	14·9 ± 0·5	*N*
(*Tvp*)		$1·56 \times 10^{-3}$	66·5	12·5 ± 0·3	

Lothidok are unconformably overlain by basalts which are considered, on petrological grounds, to be equivalent to the Tvb_2 basalts in the north. However, palaeontological evidence indicates that the Lothidok basalts are much younger, and this discrepancy remains to be resolved.

The disagreement between samples 260/61 and the apparently underlying sample 264/65 might be due to loss of argon in sample 264/65, which shows a high chlorite content, or to local repetition of the succession by faulting: a similar wide scatter has been observed in five dates[6] (30–69 m. yr.) for the Abbai basalts of Ethiopia. The age and character of the Turkana basalts are comparable with those of the (Upper) Magdala Group of the Ethiopian trap series (25 and 33 m.yr.).

The lava succession is best seen in the Lokitaung gorge, where 5,000 ft. of lavas, pyroclastics and sediments are exposed. The complex faulting in the vicinity of Lokitaung has also complicated the sequence of magnetic polarities previously described in the Lokitaung section[8]. The samples of the dated lavas appear to be magnetically stable. The results of thermal and alternating-field cleaning were found to agree closely (Fig. 1), and the shapes of the alternating-field demagnetization curves of the saturation remanence and of the natural remanence were similar (Fig. 2). High-field heating showed only single curie points—a value of 450°–500° C for the basalts, and 150°–300° C for the phonolites and nephelinites (*Tvp*). The last-mentioned, on heating, developed a new, strongly magnetic component with a curie point of about 500°–550° C. Preliminary examinations of the opaque minerals showed a low state of oxidation, with some maghemitization.

The polarities of the sites radiometrically dated are given in Table 1. A further site, 262/3, probably in the same flow as 260/1, had a polarity which became reversed after cleaning by either thermal or alternating-field demagnetization.

The existence of both polarities above the Lokitaung fault has been confirmed by 150 block measurements made in the field (1965) using an astatic magnetometer. No reversed polarities were observed below the fault. The

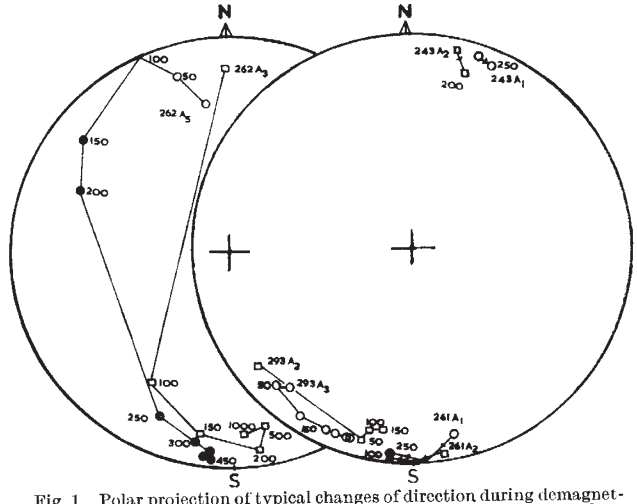


Fig. 1. Polar projection of typical changes of direction during demagnetization. ●, Dips down; ○, dips up; thermal demagnetization—numbers indicate temperatures in °C. ■, Dips down; □, dips up: alternating field demagnetization—numbers indicate peak field in oersteds

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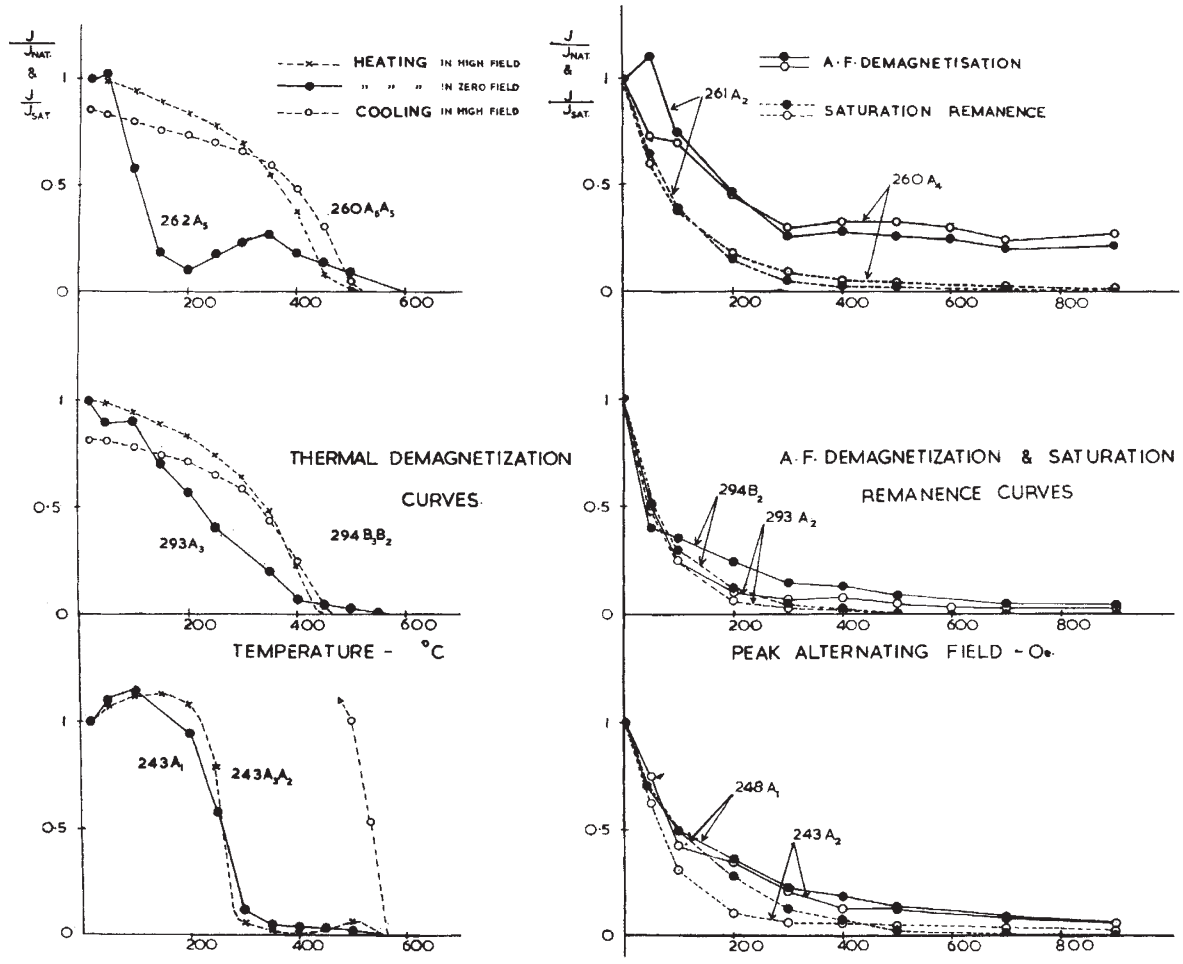


Fig. 2. Behaviour typical of dated sites during thermal and alternating-field demagnetization of natural remanent magnetism, alternating-field demagnetization of saturation remanence and high-field heating and cooling curves

reversed polarities occur in many rock types, an agglomerate horizon having a consistently reversed polarity, whereas highly mineralized fault breccias are magnetized normally.

Self-reversal and tectonic repetition are probably both present, but the foregoing observations support the previous suggestion of a field reversal in the Upper Oligocene preceded by a period of normal polarity.

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Jebel Marra Volcano, Sudan

HAMMERTON'S suggestion[1] that the volcano of Jebel Marra is dormant rather than extinct is not only of considerable vulcanological interest, but also of considerable practical importance to the future development of the area at present being investigated by the United Nations.

These investigations, started for the Sudan government by Hunting Technical Services, Limited, in the winter of 1957–58, and subsequently participated in by an ecologist and pedologists from the company for the United Nations Special Fund Jebel Marra Project, have disclosed new information regarding the formation of the volcano.

The 1957–58 geological investigations led to the recognition of three phases of volcanic activity at Jebel Marra. The first phase resulted in the accumulation of a great thickness of basalt and trachyte lavas and tuffs which form the main mass of the volcano. A period of relative quiescence ensued during which the newly formed mountain was subjected to intense sub-aerial erosion resulting in the formation of extensive piedmont alluvial and colluvial deposits. Relatively small quantities of basaltic lavas, intercalated in, or overlying, the alluvial deposits, represent the second volcanic phase. The third and final phase was entirely explosive and resulted in the formation of extensive deposits of tuff.

No new evidence regarding the age of volcanic activity was obtained in 1957–58. Since then White[2] has made observations which suggest that the ash eruptions may have continued into recent, possibly early historical, times.

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